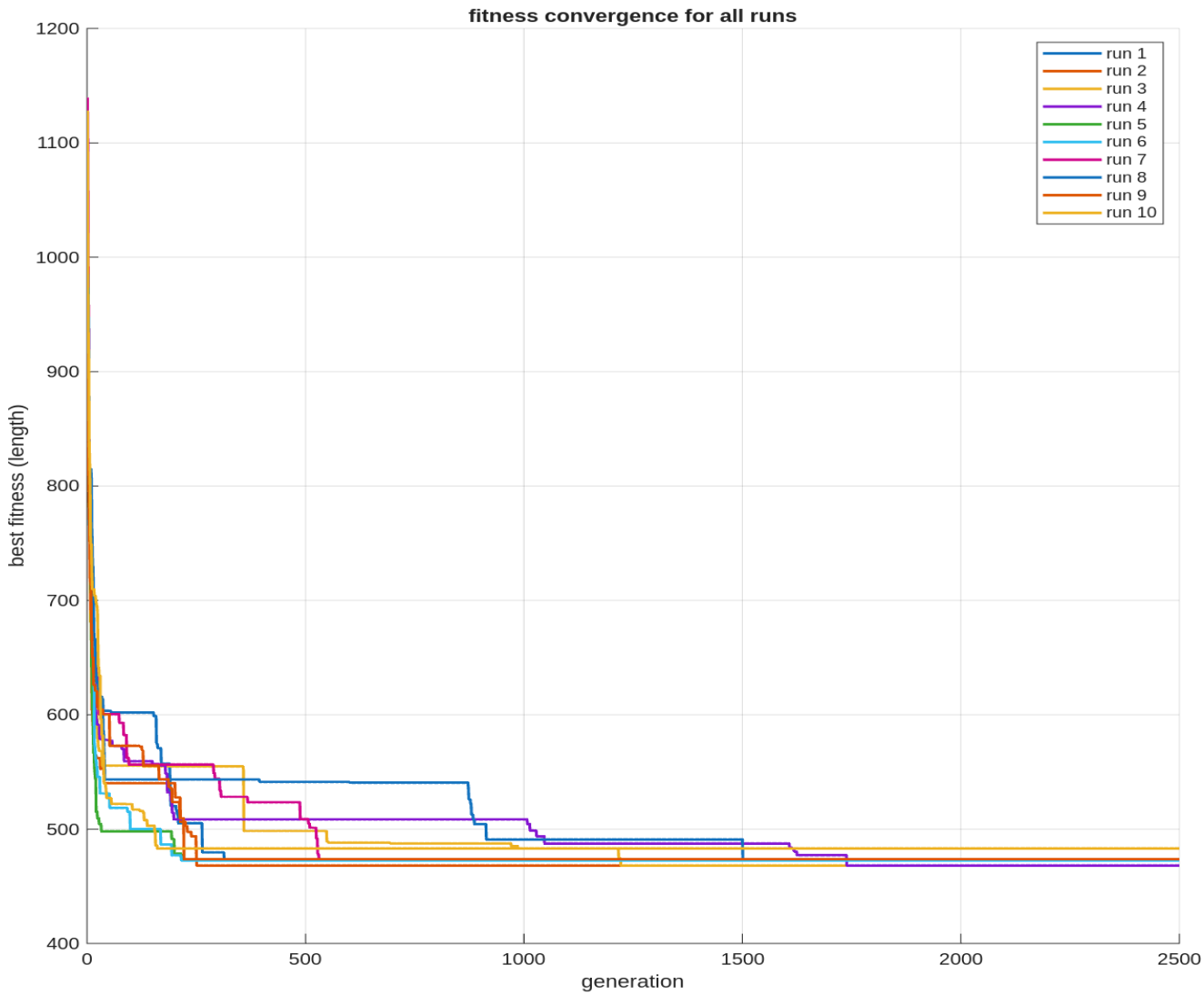
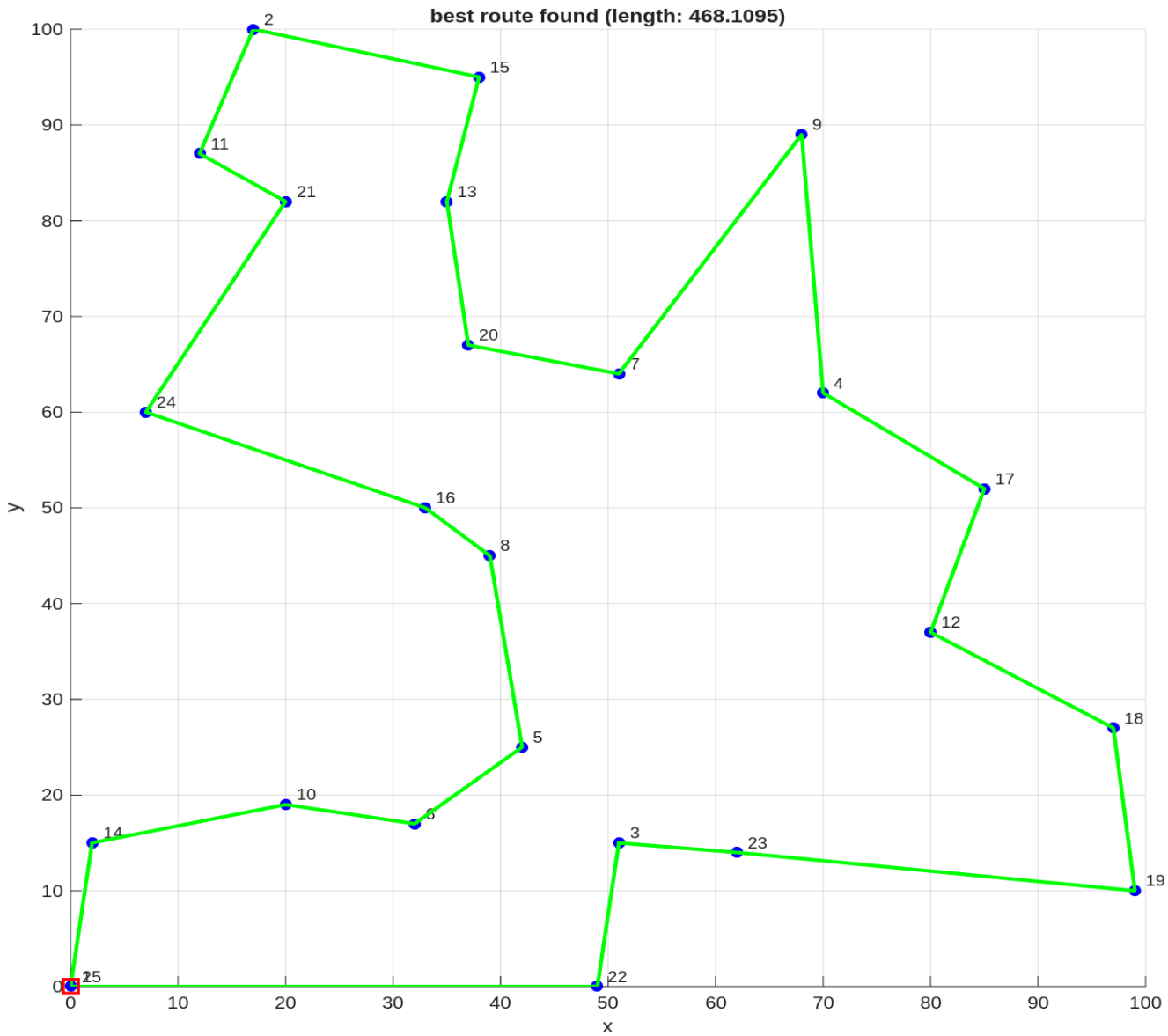
Umint cv3

Best



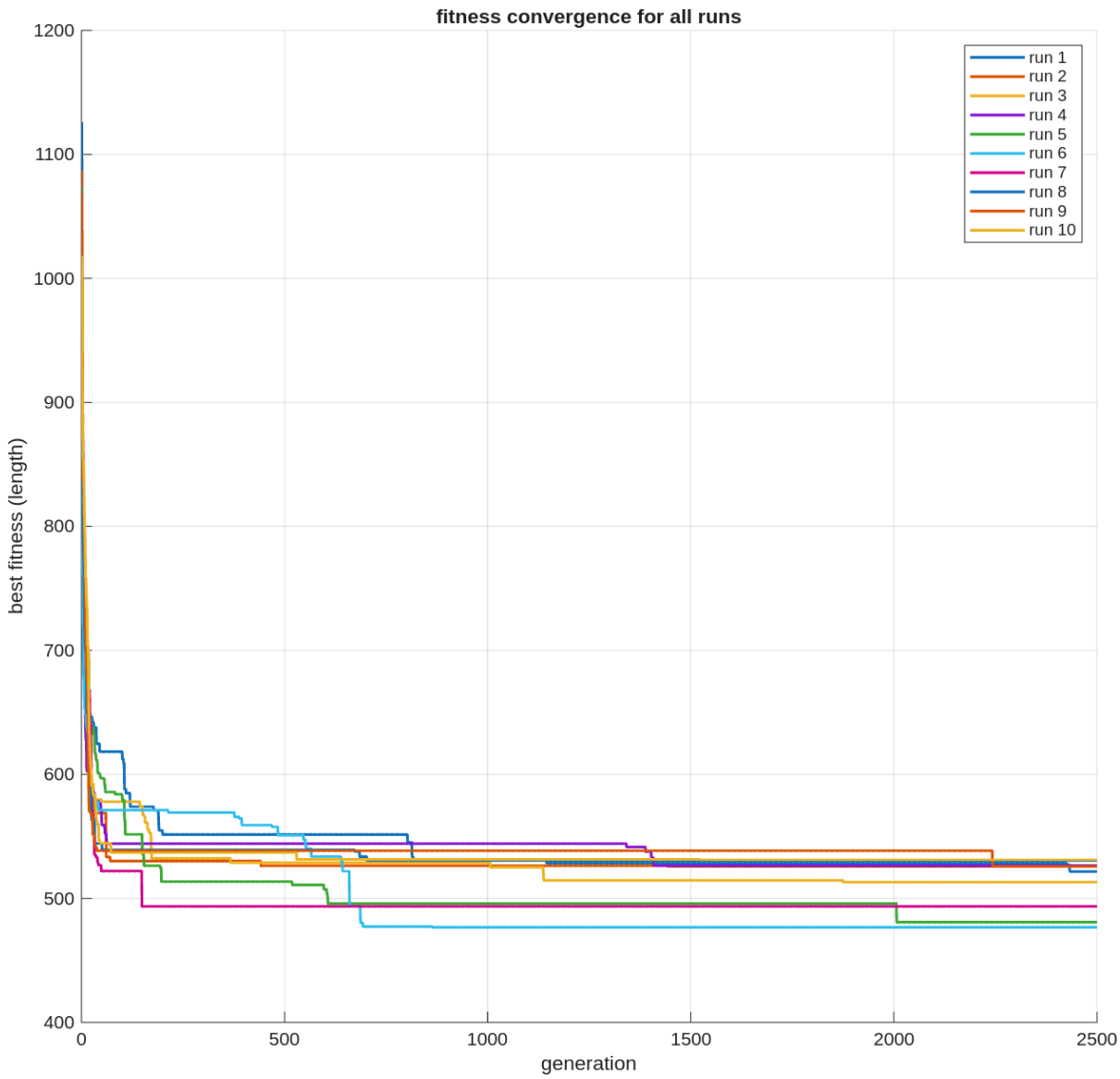


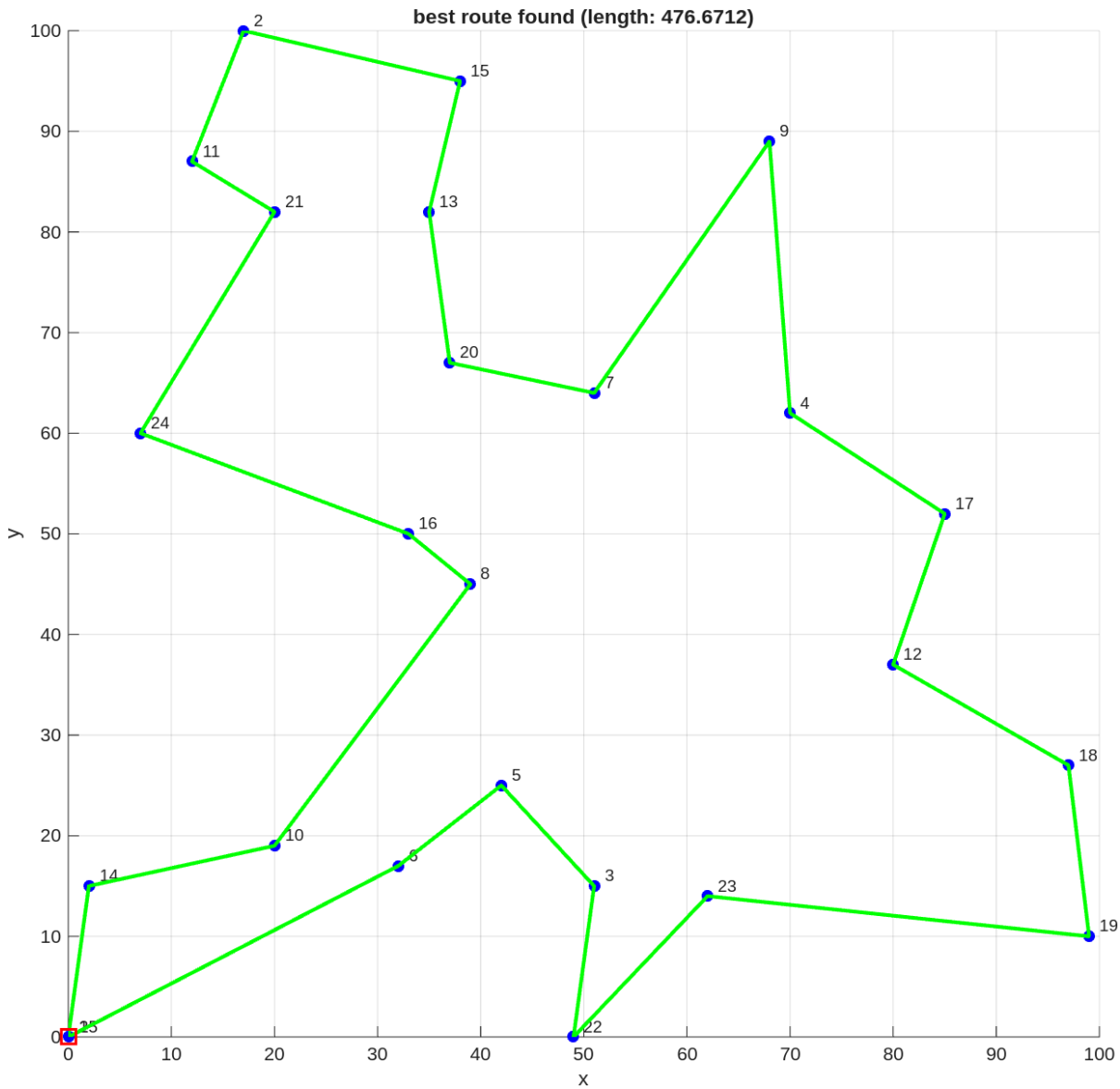

```
popSize = 100;
elitismCount = 5;
mutationRate = 0.12;
numGenerations = 2500;
numRuns = 10;


starting calculations (10 runs)...
run 1 finished. best fitness: 473.7614
run 2 finished. best fitness: 468.1095
run 3 finished. best fitness: 468.1095
run 4 finished. best fitness: 468.1095
run 5 finished. best fitness: 472.5849
run 6 finished. best fitness: 472.5849
run 7 finished. best fitness: 473.7614
run 8 finished. best fitness: 473.7614
run 9 finished. best fitness: 473.7614
run 10 finished. best fitness: 483.1013
```

```
==================== results ====================
absolute best route length: 468.1095
number of successful runs (<= 480): 9 of 10
best route sequence (point indices):
1 22 3 23 19 18 12 17 4 9 7 20 13 15 2 11 21 24 16 8 5 6 10 14 25
```

Without invord

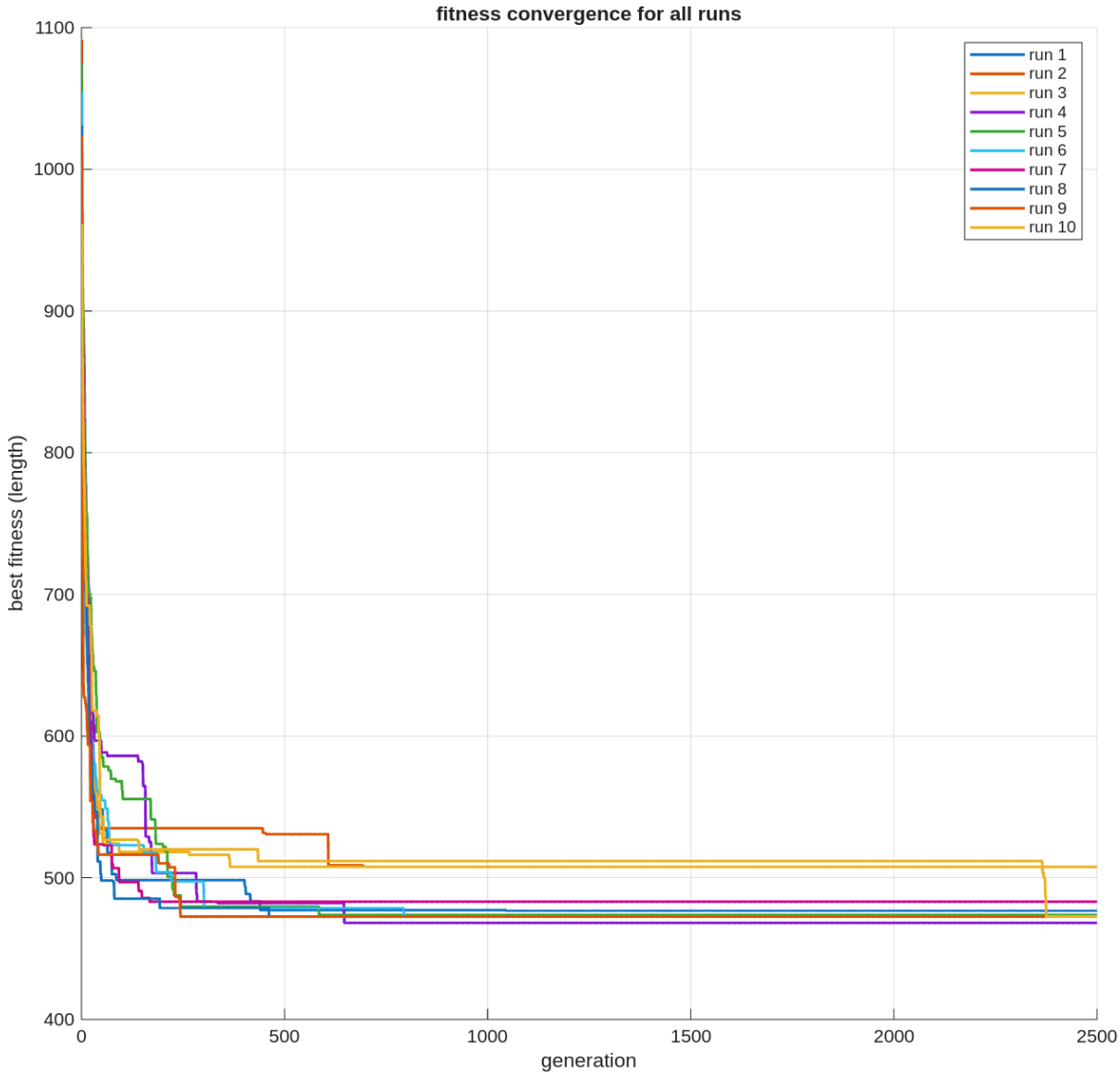


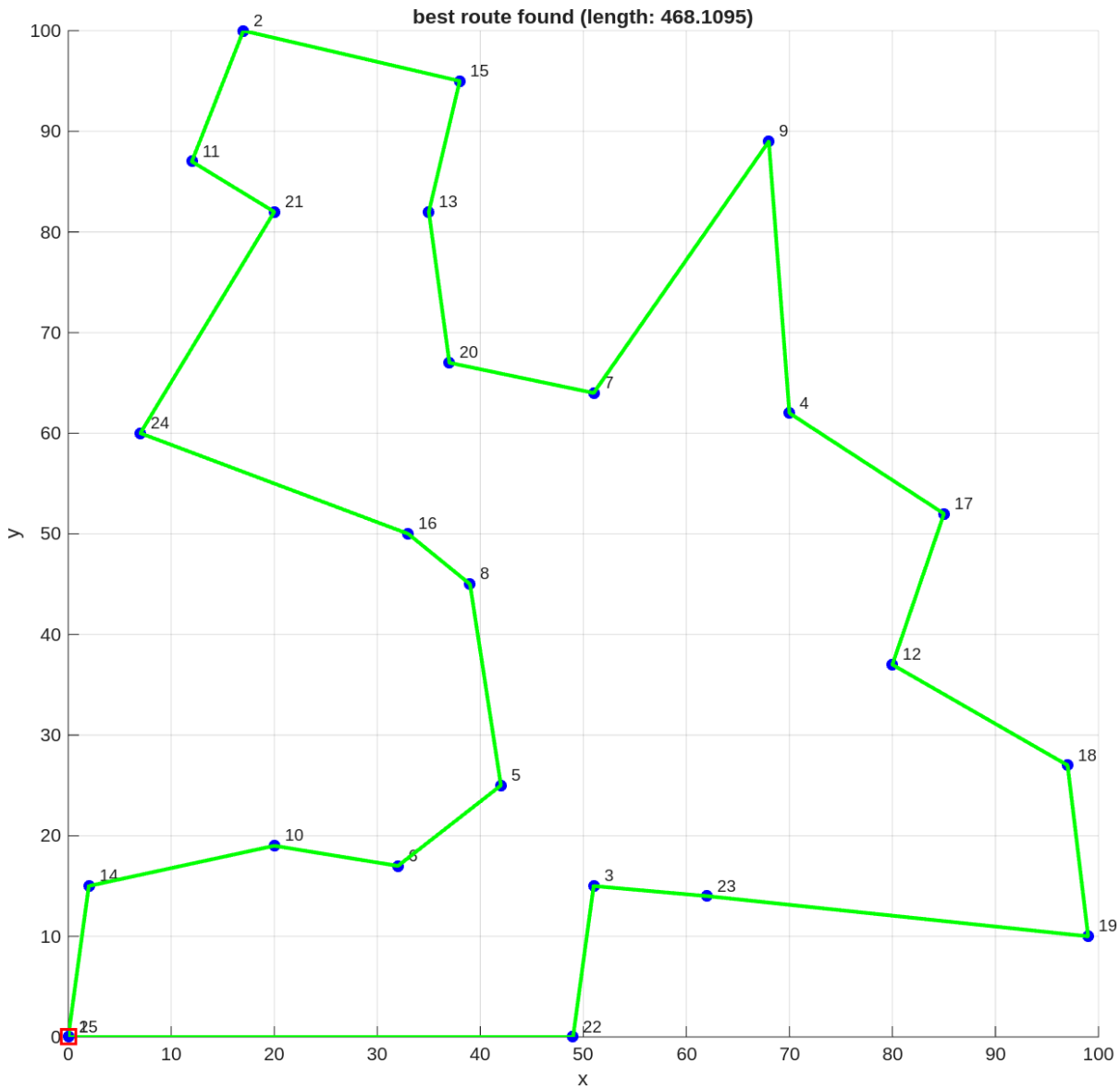



```
popSize = 100;
elitismCount = 5;
mutationRate = 0.12;
numGenerations = 2500;
numRuns = 10;
starting calculations (10 runs)...
run 1 finished. best fitness: 521.7054
run 2 finished. best fitness: 525.7512
run 3 finished. best fitness: 513.1321
run 4 finished. best fitness: 526.4064
run 5 finished. best fitness: 480.7887
run 6 finished. best fitness: 476.6712
run 7 finished. best fitness: 493.4724
run 8 finished. best fitness: 530.4446
run 9 finished. best fitness: 525.7512
run 10 finished. best fitness: 531.0928

==================== results ====================
absolute best route length: 476.671241
number of successful runs (<= 480): 1 of 10
```

Without swappart

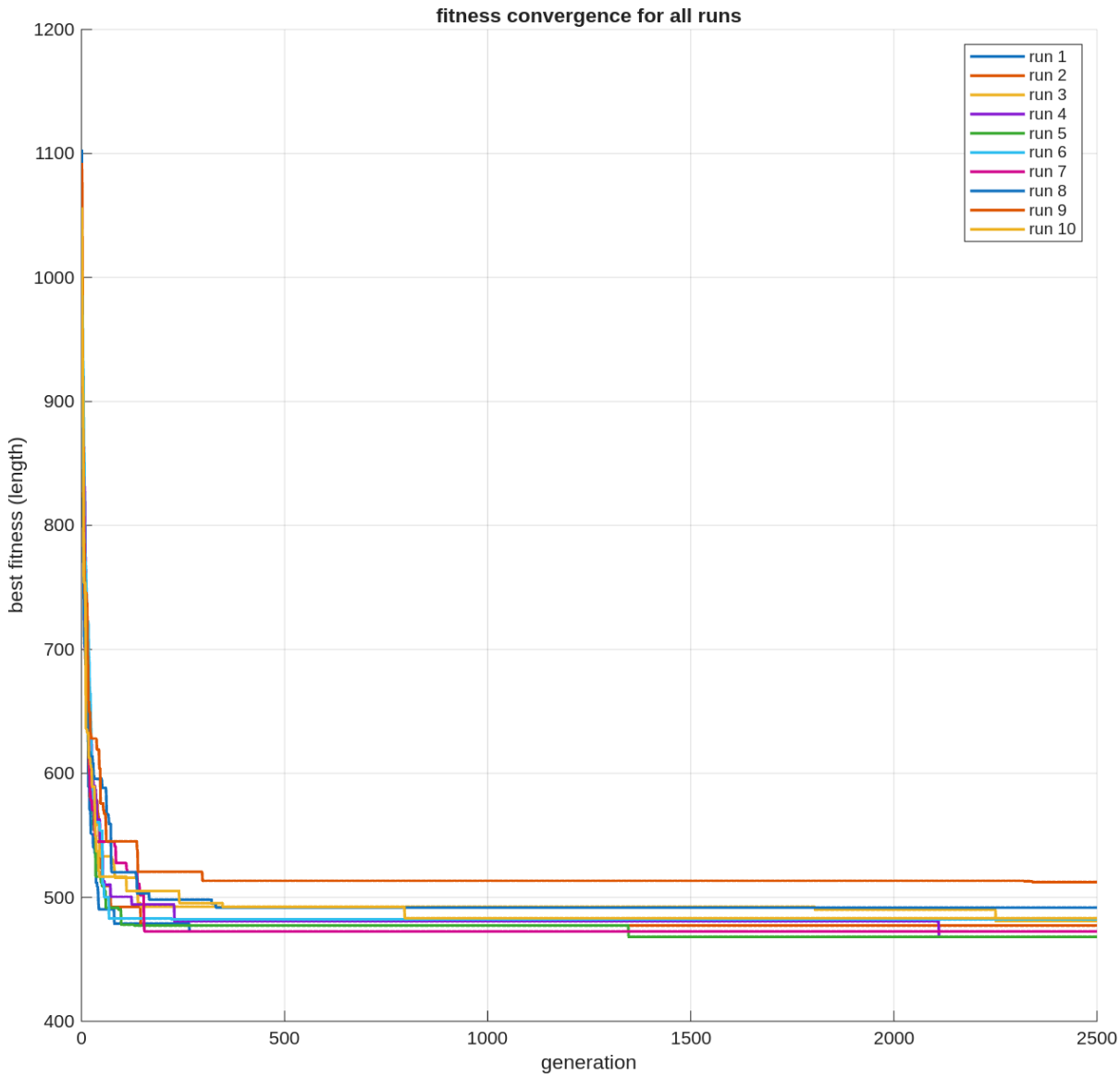


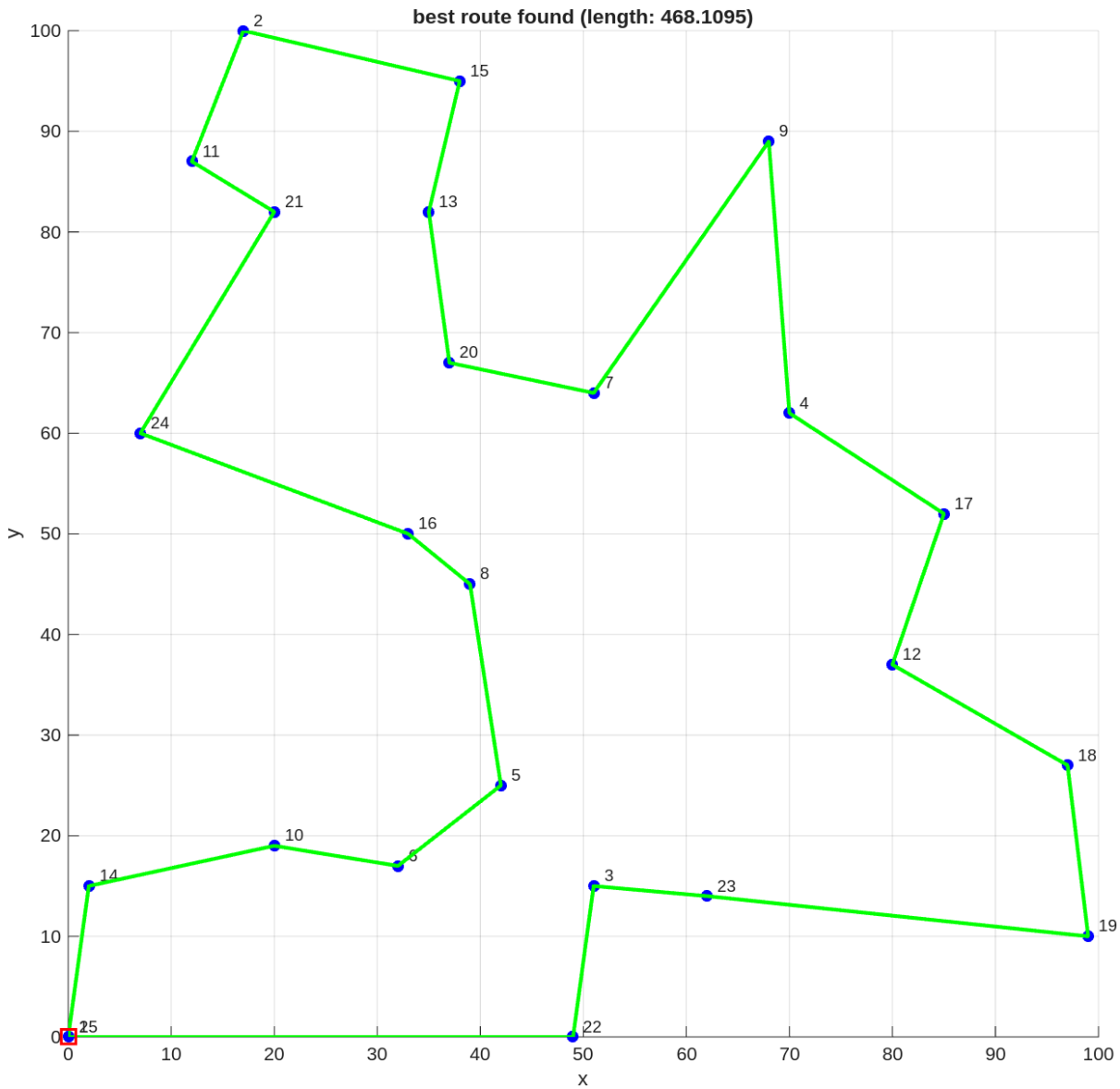



```
popSize = 100;
elitismCount = 5;
mutationRate = 0.12;
numGenerations = 2500;
numRuns = 10;
starting calculations (10 runs)...
run 1 finished. best fitness: 476.6712
run 2 finished. best fitness: 507.7905
run 3 finished. best fitness: 507.7905
run 4 finished. best fitness: 468.1095
run 5 finished. best fitness: 473.7614
run 6 finished. best fitness: 472.5849
run 7 finished. best fitness: 483.1013
run 8 finished. best fitness: 472.5849
run 9 finished. best fitness: 472.5849
run 10 finished. best fitness: 472.5849

==================== results ====================
absolute best route length: 468.109452
number of successful runs (<= 480): 7 of 10
```

Without swapgen



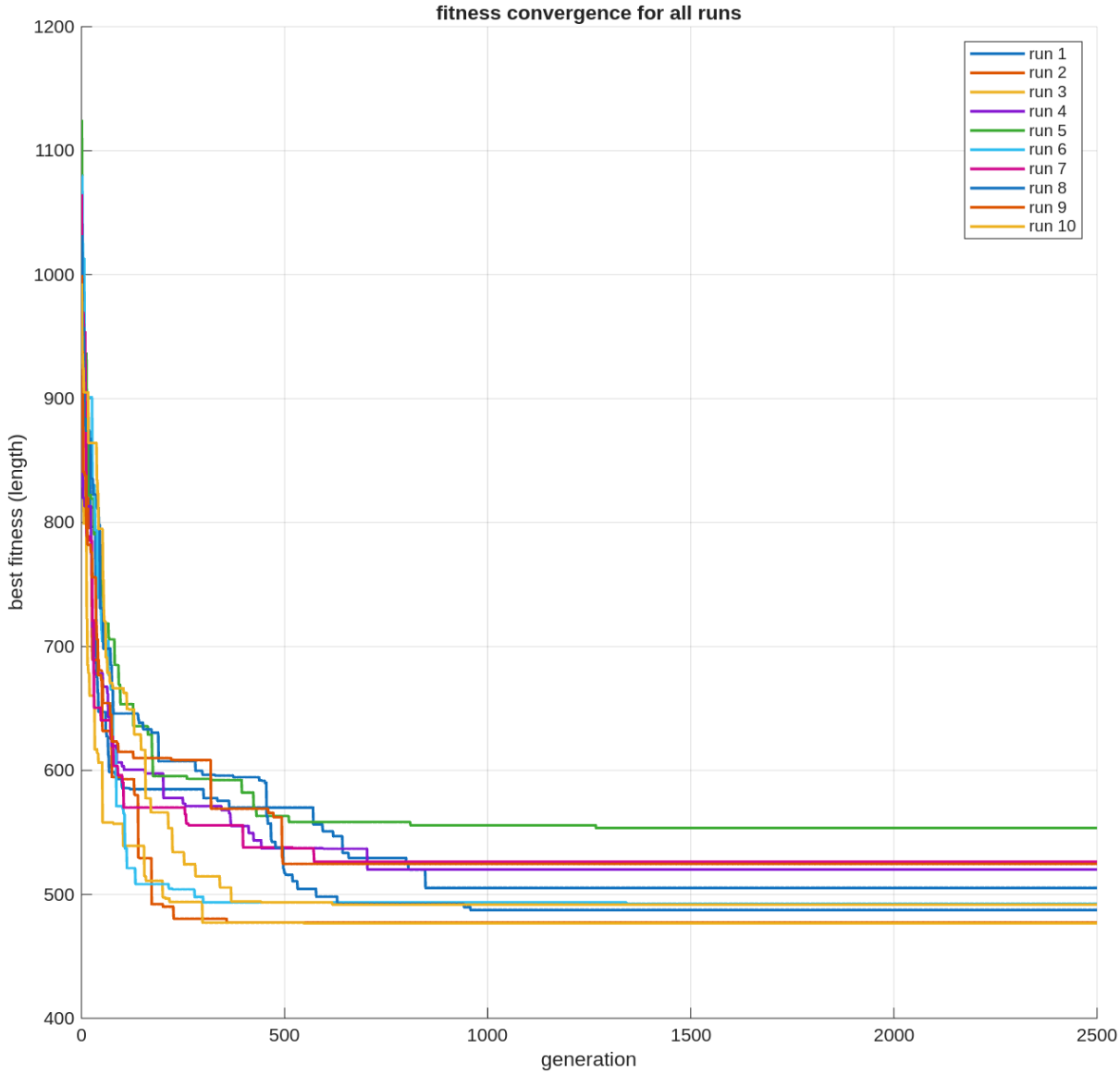


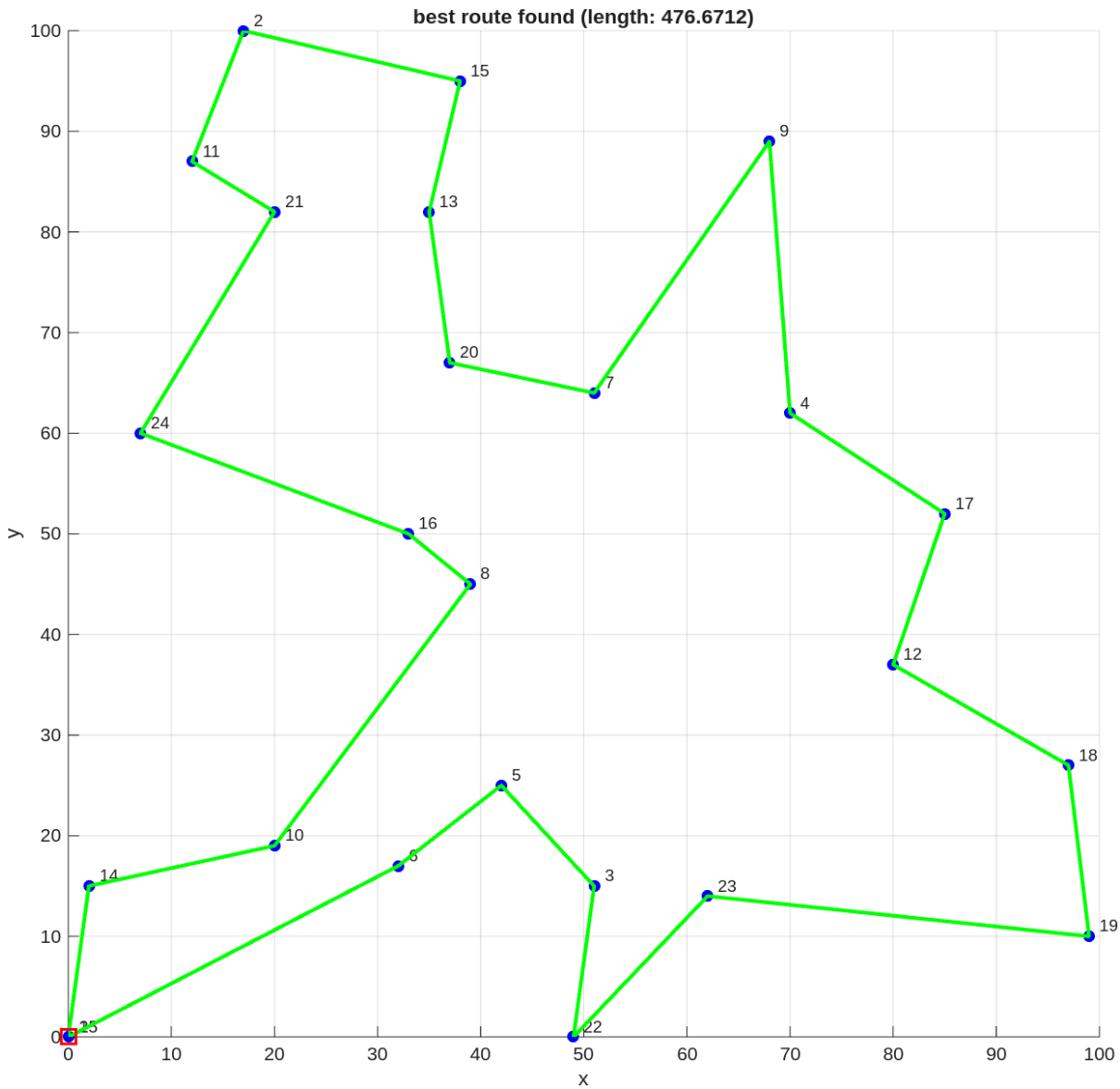

```
popSize = 100;
elitismCount = 5;
mutationRate = 0.12;
numGenerations = 2500;
numRuns = 10;
starting calculations (10 runs)...
run 1 finished. best fitness: 472.5849
run 2 finished. best fitness: 477.2182
run 3 finished. best fitness: 480.8947
run 4 finished. best fitness: 468.1095
run 5 finished. best fitness: 468.1095
run 6 finished. best fitness: 482.2311
run 7 finished. best fitness: 472.5849
run 8 finished. best fitness: 491.7852
run 9 finished. best fitness: 512.3043
run 10 finished. best fitness: 483.1554

==================== results ====================
```

```
absolute best route length: 468.109452
number of successful runs (<= 480): 5 of 10
```

MutRate = 0.9

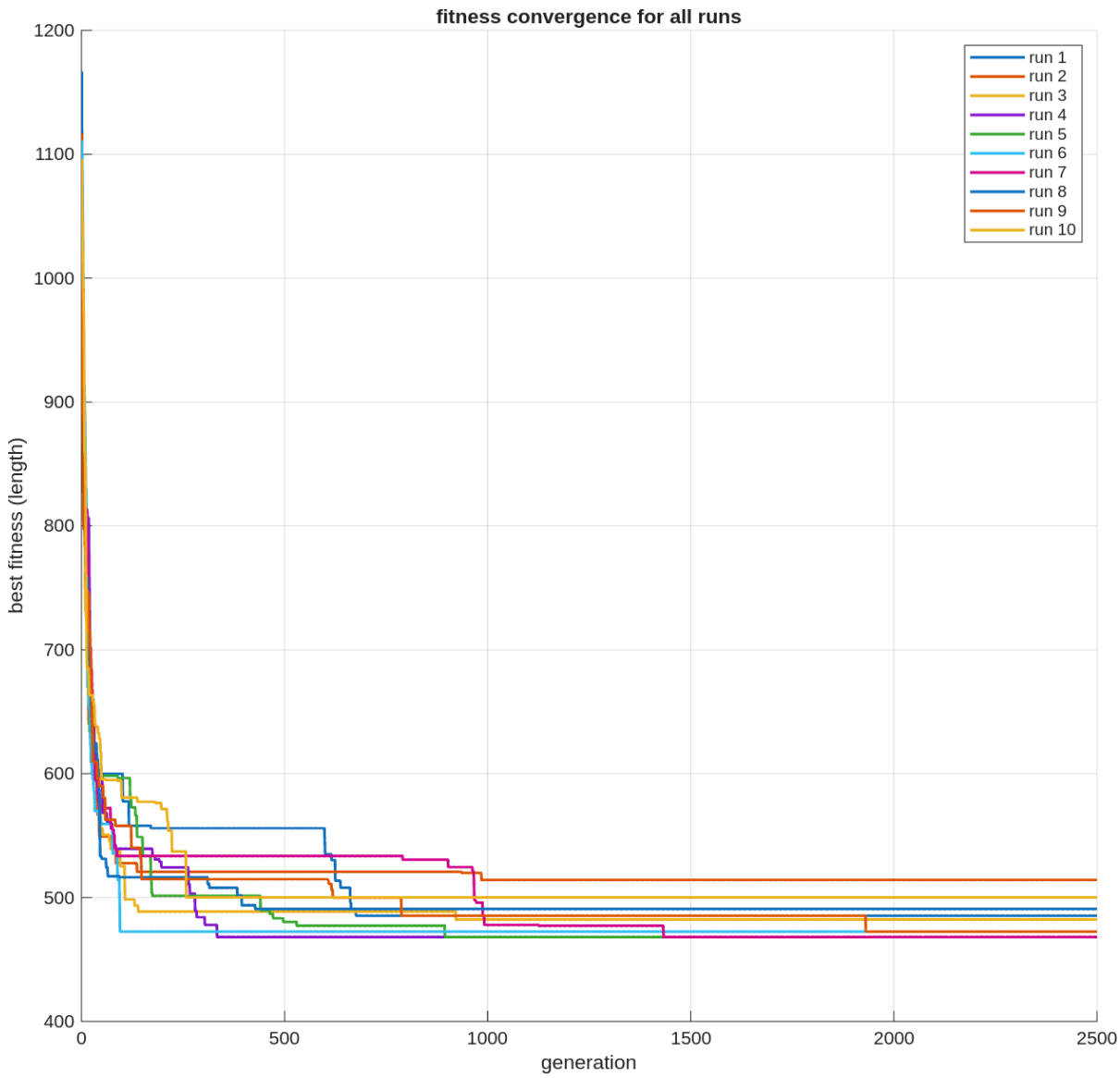


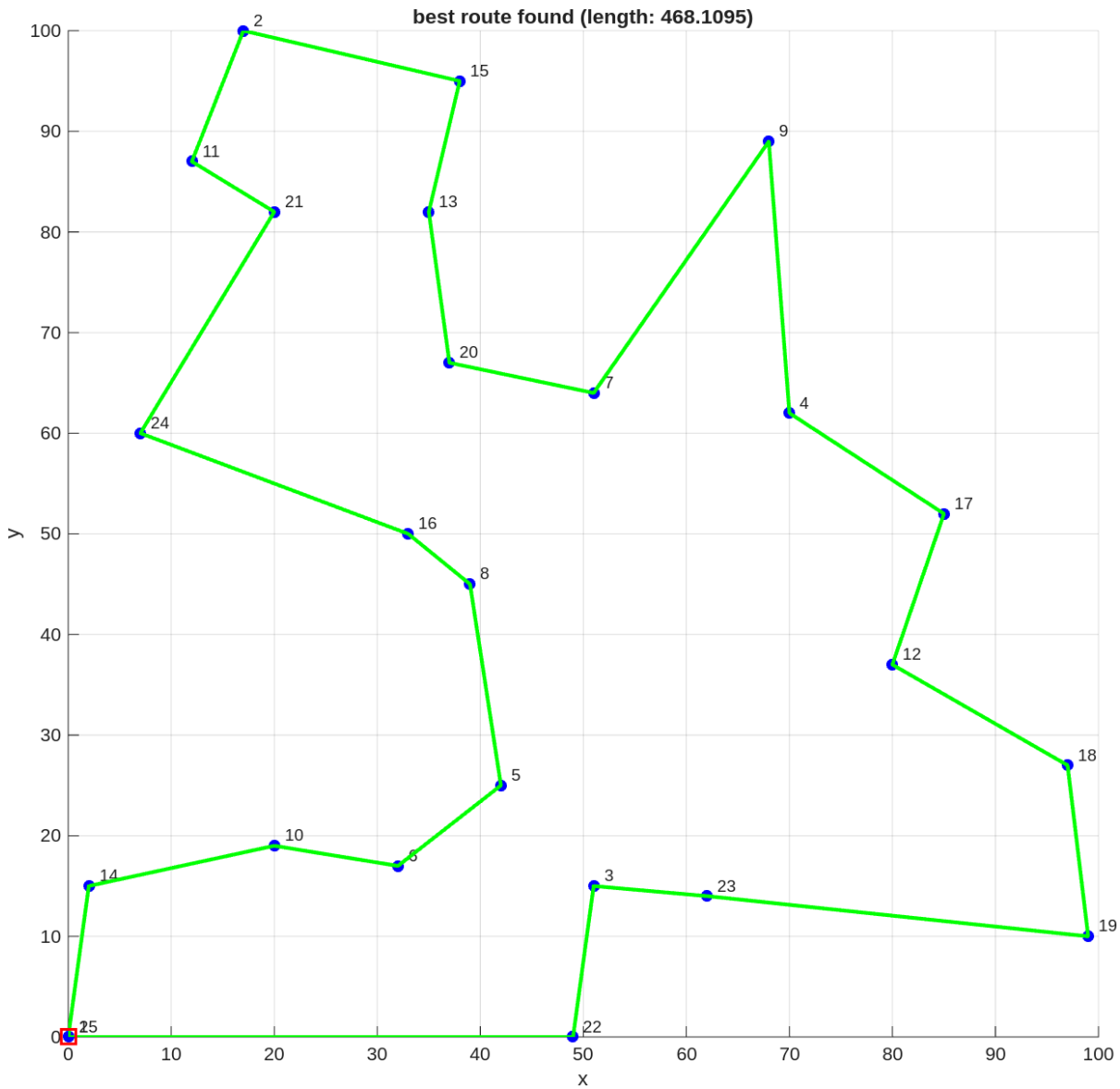



```
popSize = 100;
elitismCount = 5;
mutationRate = 0.9;
numGenerations = 2500;
numRuns = 10;
starting calculations (10 runs)...
run 1 finished. best fitness: 505.1680
run 2 finished. best fitness: 477.2182
run 3 finished. best fitness: 476.6712
run 4 finished. best fitness: 520.0965
run 5 finished. best fitness: 553.5806
run 6 finished. best fitness: 492.4358
run 7 finished. best fitness: 526.1795
run 8 finished. best fitness: 487.3989
run 9 finished. best fitness: 524.6329
run 10 finished. best fitness: 491.7172

==================== results ====================
absolute best route length: 476.671241
number of successful runs (<= 480): 2 of 10
```

```
PopSize = 50
```

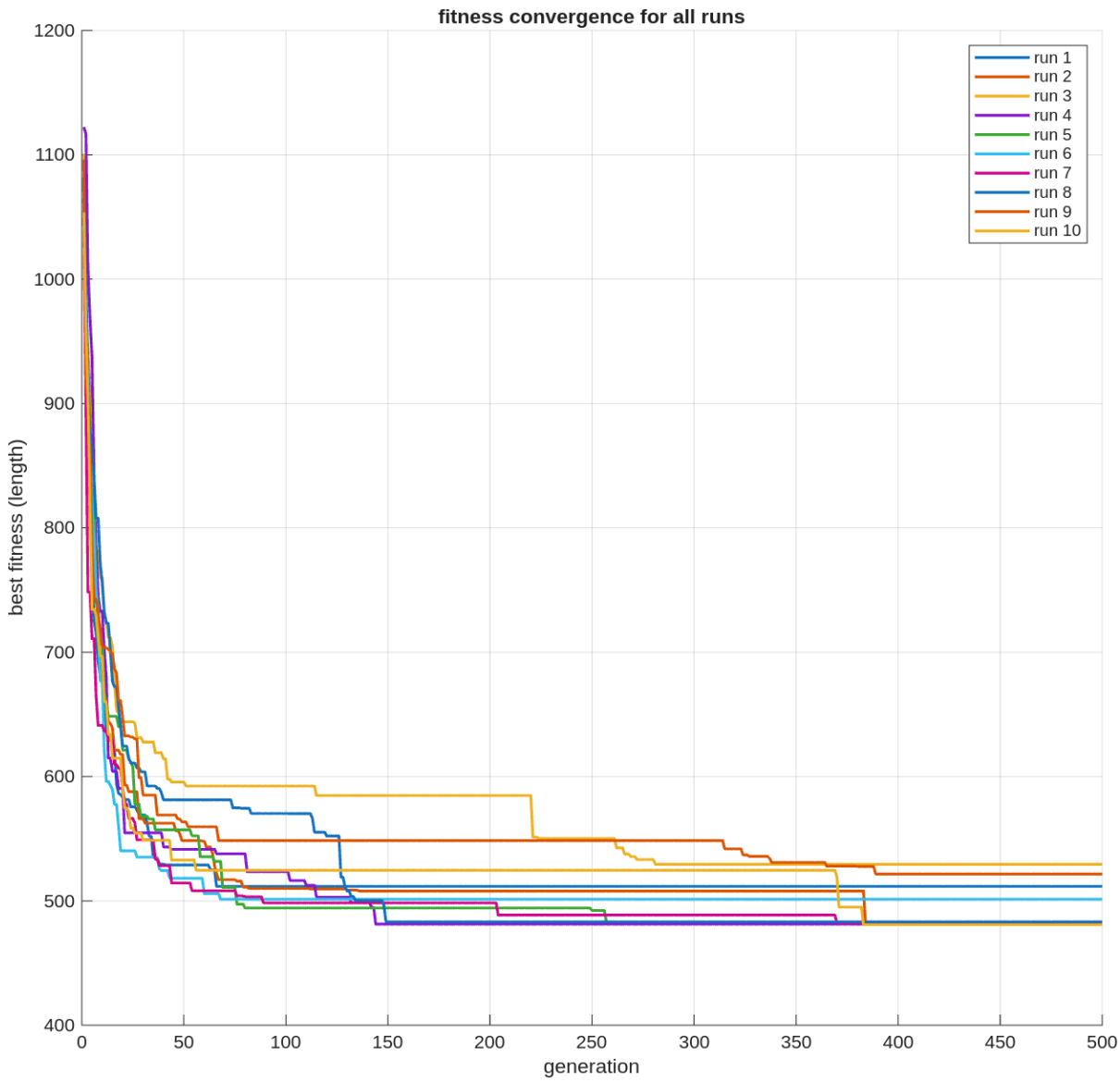

fitness convergence for all runs
best fitness (length)
generation
1200
1100
1000
900
800
700
600
500
400
0
500
1000
1500
2000
2500
run 1
run 2
run 3
run 4
run 5
run 6
run 7
run 8
run 9
run 10

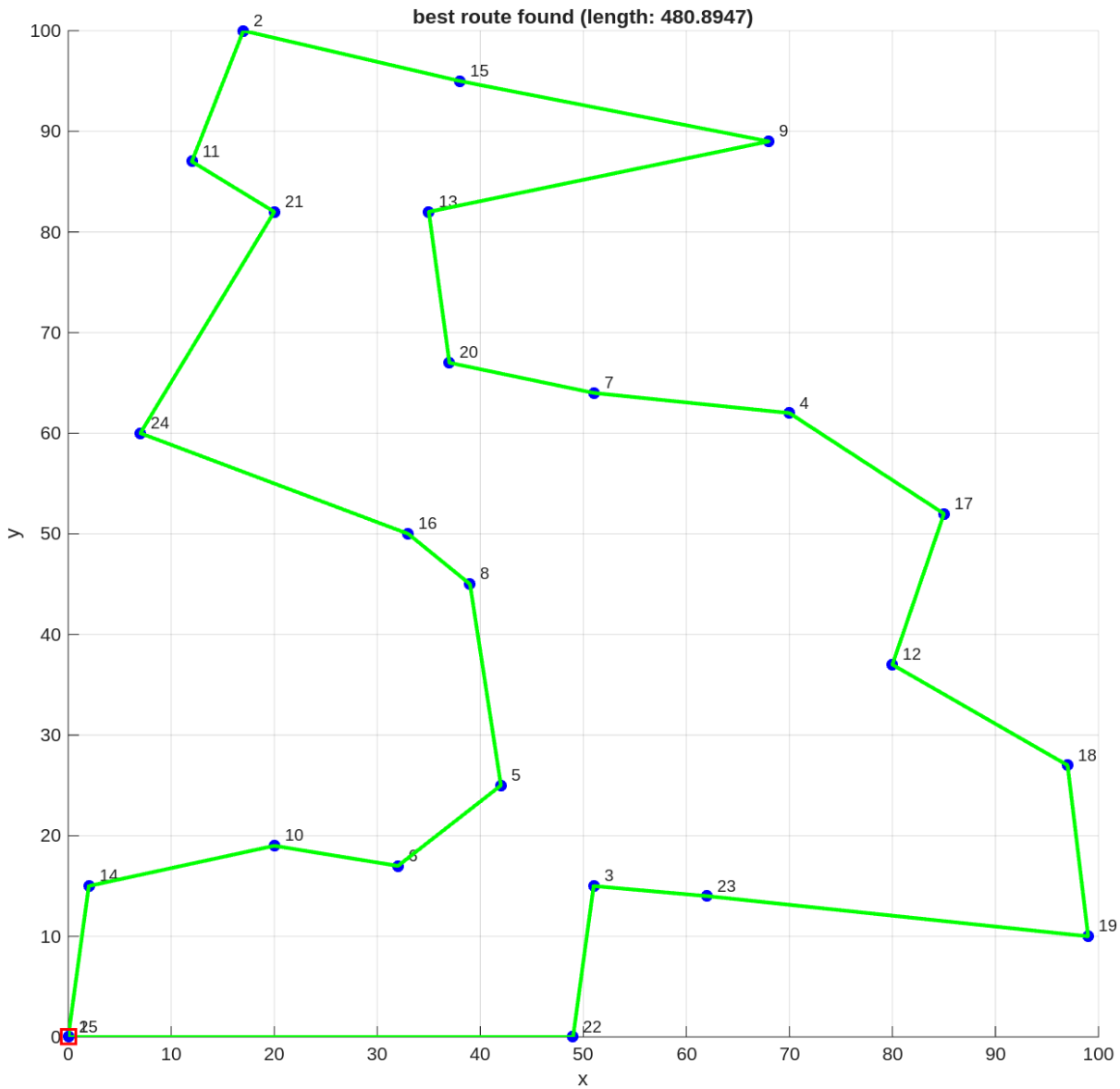



```
popSize = 50;
elitismCount = 5;
mutationRate = 0.12;
numGenerations = 2500;
numRuns = 10;
starting calculations (10 runs)...
run 1 finished. best fitness: 485.3701
run 2 finished. best fitness: 514.1704
run 3 finished. best fitness: 482.2311
run 4 finished. best fitness: 468.1095
run 5 finished. best fitness: 468.1095
run 6 finished. best fitness: 472.5849
run 7 finished. best fitness: 468.1095
run 8 finished. best fitness: 490.7929
run 9 finished. best fitness: 472.5849
run 10 finished. best fitness: 500.0835

==================== results ====================
absolute best route length: 468.109452
number of successful runs (<= 480): 5 of 10
```

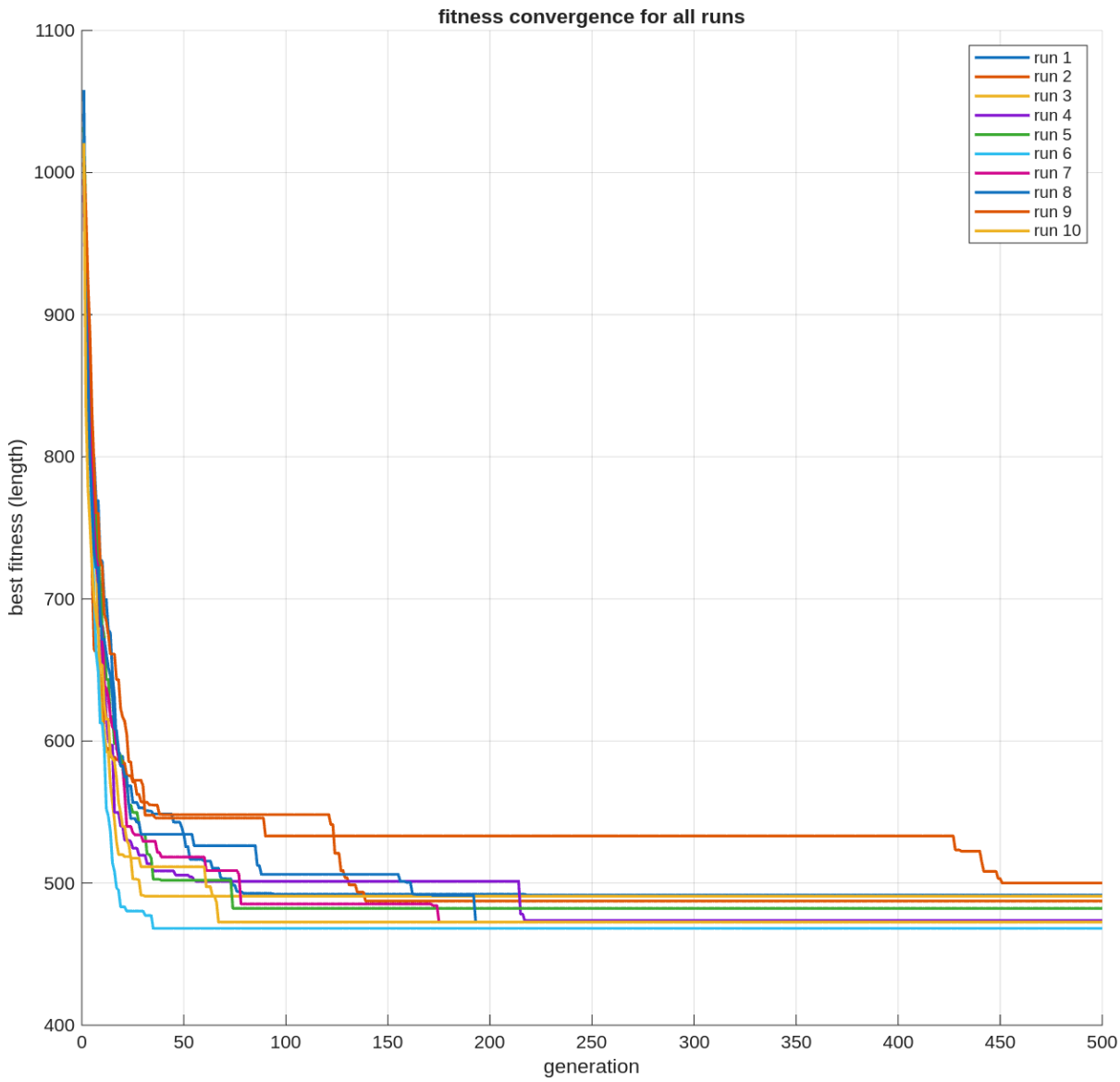
NumGeneration = 500

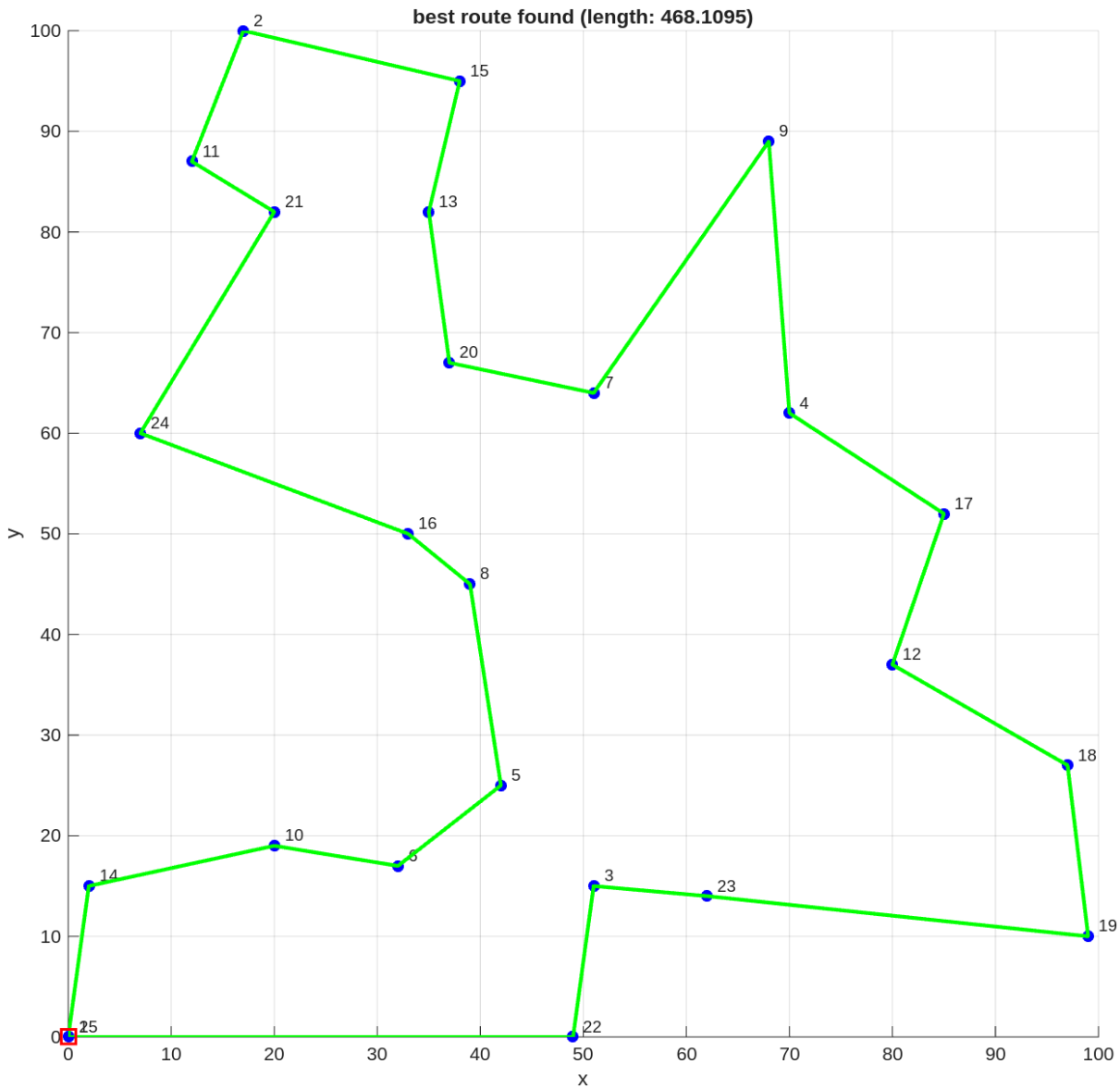

fitness convergence for all runs
best fitness (length)
generation
1200
1100
1000
900
800
700
600
500
400
0
50
100
150
200
250
300
350
400
450
500
run 1
run 2
run 3
run 4
run 5
run 6
run 7
run 8
run 9
run 10




```
popSize = 100;
elitismCount = 5;
mutationRate = 0.12;
numGenerations = 500;
numRuns = 10;
starting calculations (10 runs)...
run 1 finished. best fitness: 511.6636
run 2 finished. best fitness: 482.7820
run 3 finished. best fitness: 529.4296
run 4 finished. best fitness: 481.3255
run 5 finished. best fitness: 483.1554
run 6 finished. best fitness: 501.3852
run 7 finished. best fitness: 482.2311
run 8 finished. best fitness: 483.1554
run 9 finished. best fitness: 521.6045
run 10 finished. best fitness: 480.8947

==================== results ====================
absolute best route length: 480.894729
number of successful runs (<= 480): 0 of 10
```


fitness convergence for all runs
best fitness (length)
generation
1100
1000
900
800
700
600
500
400
0
50
100
150
200
250
300
350
400
450
500
run 1
run 2
run 3
run 4
run 5
run 6
run 7
run 8
run 9
run 10




```
popSize = 500;
elitismCount = 5;
mutationRate = 0.12;
numGenerations = 500;
numRuns = 10;
starting calculations (10 runs)...
run 1 finished. best fitness: 491.6631
run 2 finished. best fitness: 487.3989
run 3 finished. best fitness: 490.7929
run 4 finished. best fitness: 473.7614
run 5 finished. best fitness: 482.2311
run 6 finished. best fitness: 468.1095
run 7 finished. best fitness: 472.5849
run 8 finished. best fitness: 472.5849
run 9 finished. best fitness: 500.0835
run 10 finished. best fitness: 472.5849

==================== results ====================
absolute best route length: 468.109452
number of successful runs (<= 480): 5 of 10
```